# FYS4480 - Midterm Exam 1

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All results derived by hand, and all code used for the numerical parts are available in this GitHub repository. The handwritten notes can be found as PDFs and the code are python files.

## Part 1: Symmetries and matrix elements

### Task 1: Commutations

$$
\begin{aligned}
\hat{S}_z &= \frac{1}{2}\sum_{p\sigma} \sigma\, a^\dagger_{p\sigma} a_{p\sigma}, \\
\hat{S}^2 &= \hat{S}_z^2 + \frac{1}{2}\left(\hat{S}_+\hat{S}_- + \hat{S}_-\hat{S}_+\right), \\
\hat{S}_\pm &= \sum_{p} a^\dagger_{p\pm} a_{p\mp}, \\
\hat{H}_0 &= \xi \sum_{p\sigma} (p-1)\, a^\dagger_{p\sigma} a_{p\sigma}, \qquad (\xi = 1), \\
\hat{V} &= -\frac{1}{2}\, g \sum_{pq} a^\dagger_{p+} a^\dagger_{p-} a_{q-} a_{q+} = -\frac{1}{2}\, g \sum_{pq} \hat{P}^+_p \hat{P}^-_q,
\end{aligned}
$$

with the pair operators

$$
\hat{P}^+_p = a^\dagger_{p+} a^\dagger_{p-}, \qquad \hat{P}^-_q = a_{q-} a_{q+}.
$$

**(1)** $\left[\hat{H}_0, \hat{S}_z\right] = 0$

We show $\hat{H}_0\hat{S}_z = \hat{S}_z\hat{H}_0$. For fermions, number operators commute, $\left[a^\dagger_i a_i, a^\dagger_j a_j\right] = 0$. Then

$$
\begin{aligned}
\hat{H}_0\hat{S}_z &= \left[\sum_{p\sigma}(p-1)\, a^\dagger_{p\sigma} a_{p\sigma}\right]\left[\frac{1}{2}\sum_{p'\sigma'} \sigma'\, a^\dagger_{p'\sigma'} a_{p'\sigma'}\right] \\
&= \frac{1}{2}\sum_{\substack{p\sigma \\ p'\sigma'}} \sigma'(p-1)\, a^\dagger_{p\sigma} a_{p\sigma} a^\dagger_{p'\sigma'} a_{p'\sigma'} \\
&= \frac{1}{2}\sum_{\substack{p\sigma \\ p'\sigma'}} \sigma'(p-1)\, a^\dagger_{p'\sigma'} a_{p'\sigma'} a^\dagger_{p\sigma} a_{p\sigma} \\
&= \left[\frac{1}{2}\sum_{p'\sigma'} \sigma'\, a^\dagger_{p'\sigma'} a_{p'\sigma'}\right]\left[\sum_{p\sigma}(p-1)\, a^\dagger_{p\sigma} a_{p\sigma}\right] \\
&= \hat{S}_z\hat{H}_0. \qquad \blacksquare
\end{aligned}
$$

**(2)** $\left[\hat{H}_0, \hat{S}_\pm\right] = 0$ **and** $\left[\hat{H}_0, \hat{S}^2\right] = 0$

$\hat{H}_0\hat{S}_\pm = \hat{S}_\pm\hat{H}_0$**.** We move $a^\dagger_{p\sigma}a_{p\sigma}$ to the right of $a^\dagger_{p'\pm}a_{p'\mp}$ using the anticommutation relations. Here $\delta_{\sigma,\pm} = 1$ if $\sigma$ equals the sign appearing in $\hat{S}_\pm$, and $\delta_{\sigma,\mp} = 1$ if $\sigma$ equals the opposite sign. The Kronecker deltas are not zero for $p = p'$, so we keep them:

$$\begin{aligned}
a^\dagger_{p\sigma}a_{p\sigma}a^\dagger_{p'\pm}a_{p'\mp} &= a^\dagger_{p\sigma}\left(\delta_{pp'}\delta_{\sigma,\pm} - a^\dagger_{p'\pm}a_{p\sigma}\right)a_{p'\mp} \\
&= \delta_{pp'}\delta_{\sigma,\pm}\, a^\dagger_{p\sigma}a_{p'\mp} + a^\dagger_{p\sigma}a^\dagger_{p'\pm}a_{p'\mp}a_{p\sigma} \\
&= \delta_{pp'}\delta_{\sigma,\pm}\, a^\dagger_{p\sigma}a_{p'\mp} - a^\dagger_{p'\pm}a^\dagger_{p\sigma}a_{p'\mp}a_{p\sigma} \\
&= \delta_{pp'}\delta_{\sigma,\pm}\, a^\dagger_{p\sigma}a_{p'\mp} - a^\dagger_{p'\pm}\left(\delta_{pp'}\delta_{\sigma,\mp} - a_{p'\mp}a^\dagger_{p\sigma}\right)a_{p\sigma} \\
&= a^\dagger_{p'\pm}a_{p'\mp}a^\dagger_{p\sigma}a_{p\sigma} + \delta_{pp'}\left(\delta_{\sigma,\pm} - \delta_{\sigma,\mp}\right)a^\dagger_{p'\pm}a_{p'\mp}.
\end{aligned}$$

(In the last step $p = p'$ and $\sigma = \pm$ in the first delta term, and $\sigma = \mp$ in the second, which gives $a^\dagger_{p'\pm}a_{p'\mp}$ both times.) Inserting this in $\hat{H}_0\hat{S}_\pm$,

$$\begin{aligned}
\hat{H}_0\hat{S}_\pm &= \sum_{\substack{p\sigma \\ p'}}(p-1)\, a^\dagger_{p\sigma}a_{p\sigma}a^\dagger_{p'\pm}a_{p'\mp} \\
&= \sum_{\substack{p\sigma \\ p'}}(p-1)\, a^\dagger_{p'\pm}a_{p'\mp}a^\dagger_{p\sigma}a_{p\sigma} + \sum_{p'}(p'-1)\Big[\sum_\sigma\left(\delta_{\sigma,\pm} - \delta_{\sigma,\mp}\right)\Big]a^\dagger_{p'\pm}a_{p'\mp} \\
&= \hat{S}_\pm\hat{H}_0 + 0 \;=\; \hat{S}_\pm\hat{H}_0. \qquad \blacksquare
\end{aligned}$$

The extra terms are non-zero for each single $\sigma$, but cancel when we sum over $\sigma$, since $\sum_\sigma(\delta_{\sigma,\pm} - \delta_{\sigma,\mp}) = 1 - 1 = 0$. This works because both spins on level $p'$ carry the same weight $(p'-1)$ in $\hat{H}_0$.

$\hat{H}_0\hat{S}^2 = \hat{S}^2\hat{H}_0$**.**

$$\begin{aligned}
\hat{H}_0\hat{S}^2 &= \hat{H}_0\left[\hat{S}_z^2 + \frac{1}{2}\left(\hat{S}_+\hat{S}_- + \hat{S}_-\hat{S}_+\right)\right] \\
&= \hat{H}_0\hat{S}_z\hat{S}_z + \frac{1}{2}\hat{H}_0\hat{S}_+\hat{S}_- + \frac{1}{2}\hat{H}_0\hat{S}_-\hat{S}_+ \\
&= \hat{S}_z^2\hat{H}_0 + \frac{1}{2}\hat{S}_+\hat{S}_-\hat{H}_0 + \frac{1}{2}\hat{S}_-\hat{S}_+\hat{H}_0 \\
&= \left[\hat{S}_z^2 + \frac{1}{2}\left(\hat{S}_+\hat{S}_- + \hat{S}_-\hat{S}_+\right)\right]\hat{H}_0 \\
&= \hat{S}^2\hat{H}_0. \qquad \blacksquare
\end{aligned}$$

**(3)** $\left[\hat{V}, \hat{S}_z\right] = 0$

We first show $\left[\hat{S}_z, \hat{P}^+_p\right] = 0$. Write

$$\hat{S}_z\hat{P}^+_{p'} = \frac{1}{2}\sum_{p\sigma}\sigma\, a^\dagger_{p\sigma}a_{p\sigma}a^\dagger_{p'+}a^\dagger_{p'-}.$$

Anticommuting $a_{p\sigma}$ through to the right,

$$\begin{aligned}
a^\dagger_{p\sigma}a_{p\sigma}a^\dagger_{p'+}a^\dagger_{p'-} &= a^\dagger_{p\sigma}\left(\delta_{pp'}\delta_{\sigma+} - a^\dagger_{p'+}a_{p\sigma}\right)a^\dagger_{p'-} \\
&= a^\dagger_{p\sigma}a^\dagger_{p'-}\,\delta_{pp'}\delta_{\sigma+} - a^\dagger_{p\sigma}a^\dagger_{p'+}\left(\delta_{pp'}\delta_{\sigma-} - a^\dagger_{p'-}a_{p\sigma}\right) \\
&= a^\dagger_{p\sigma}a^\dagger_{p'-}\,\delta_{pp'}\delta_{\sigma+} - a^\dagger_{p\sigma}a^\dagger_{p'+}\,\delta_{pp'}\delta_{\sigma-} + a^\dagger_{p\sigma}a^\dagger_{p'+}a^\dagger_{p'-}a_{p\sigma}.
\end{aligned}$$

The last term equals $a^\dagger_{p'+}a^\dagger_{p'-}a^\dagger_{p\sigma}a_{p\sigma}$, which is the $\hat{P}^+_{p'}\hat{S}_z$ contribution. The first two terms are non-zero only for $p = p'$:

$$a^\dagger_{p'\sigma}a^\dagger_{p'-}\,\delta_{\sigma+} - a^\dagger_{p'\sigma}a^\dagger_{p'+}\,\delta_{\sigma-}.$$

Putting this back into the sum over $\sigma$ (with the factor $\sigma$):

$$\begin{aligned}\frac{1}{2}\sum_\sigma \sigma\left[a^\dagger_{p'\sigma}a^\dagger_{p'-}\,\delta_{\sigma+} - a^\dagger_{p'\sigma}a^\dagger_{p'+}\,\delta_{\sigma-}\right] &= \frac{1}{2}\left[a^\dagger_{p'+}a^\dagger_{p'-} + a^\dagger_{p'-}a^\dagger_{p'+}\right] \\ &= \frac{1}{2}\left\{a^\dagger_{p'+}, a^\dagger_{p'-}\right\} = 0 \quad \text{(anticommutation)}.\end{aligned}$$

Hence

$$\hat{S}_z\hat{P}^+_{p'} = \frac{1}{2}\sum_{p\sigma}\sigma\, a^\dagger_{p\sigma}a_{p\sigma}a^\dagger_{p'+}a^\dagger_{p'-} = \frac{1}{2}\sum_{p\sigma}\sigma\, a^\dagger_{p'+}a^\dagger_{p'-}a^\dagger_{p\sigma}a_{p\sigma} = \hat{P}^+_{p'}\hat{S}_z,$$

so $\left[\hat{S}_z, \hat{P}^+_p\right] = 0$. The same argument gives $\left[\hat{S}_z, \hat{P}^-_p\right] = 0$. Then

$$\begin{aligned}\hat{V}\hat{S}_z &= -\frac{1}{2}g\sum_{pq}\left[\hat{P}^+_p\hat{P}^-_q\right]\hat{S}_z = -\frac{1}{2}g\sum_{pq}\hat{S}_z\left[\hat{P}^+_p\hat{P}^-_q\right] \\ &= \hat{S}_z\left[-\frac{1}{2}g\sum_{pq}\hat{P}^+_p\hat{P}^-_q\right] = \hat{S}_z\hat{V}. \qquad \blacksquare\end{aligned}$$

**(4)** $\left[\hat{V}, \hat{S}^2\right] = 0$

We first show $\left[\hat{S}_\pm, \hat{P}^+_p\right] = 0$. Using $\hat{S}_\pm = \sum_p a^\dagger_{p\pm}a_{p\mp}$,

$$\begin{aligned}\hat{S}_\pm\hat{P}^+_{p'} &= \sum_p a^\dagger_{p\pm}a_{p\mp}a^\dagger_{p'+}a^\dagger_{p'-} \\ &= \sum_p a^\dagger_{p\pm}\left(\delta_{pp'}\delta_{\mp+} - a^\dagger_{p'+}a_{p\mp}\right)a^\dagger_{p'-} \\ &= \sum_p \left[a^\dagger_{p\pm}a^\dagger_{p'-}\,\delta_{pp'}\delta_{\mp+} - a^\dagger_{p\pm}a^\dagger_{p'+}\left(\delta_{pp'}\delta_{\mp-} - a^\dagger_{p'-}a_{p\mp}\right)\right] \\ &= \sum_p \left[a^\dagger_{p\pm}a^\dagger_{p'-}\,\delta_{pp'}\delta_{\mp+} - a^\dagger_{p\pm}a^\dagger_{p'+}\,\delta_{pp'}\delta_{\mp-} + a^\dagger_{p\pm}a^\dagger_{p'+}a^\dagger_{p'-}a_{p\mp}\right] \\ &= \left[a^\dagger_{p'\pm}a^\dagger_{p'-}\,\delta_{\mp+} - a^\dagger_{p'\pm}a^\dagger_{p'+}\,\delta_{\mp-} + a^\dagger_{p\pm}a^\dagger_{p'+}a^\dagger_{p'-}a_{p\mp}\right] \text{ (summed over } p).\end{aligned}$$

The first two terms are always zero: for each sign choice either the Kronecker delta vanishes, or we get $a^\dagger_{p'-}a^\dagger_{p'-} = 0$ or $a^\dagger_{p'+}a^\dagger_{p'+} = 0$. What remains is

$$\hat{S}_\pm\hat{P}^+_{p'} = a^\dagger_{p'+}a^\dagger_{p'-}\sum_p a^\dagger_{p\pm}a_{p\mp} = \hat{P}^+_{p'}\hat{S}_\pm,$$

so $\left[\hat{S}_\pm, \hat{P}^+_p\right] = 0$. The same argument gives $\left[\hat{S}_\pm, \hat{P}^-_p\right] = 0$.

Since $\hat{S}_+$, $\hat{S}_-$ and $\hat{S}_z$ all commute with $\hat{P}^+_p$ and $\hat{P}^-_q$, so does $\hat{S}^2 = \hat{S}^2_z + \frac{1}{2}(\hat{S}_+\hat{S}_- + \hat{S}_-\hat{S}_+)$:

$$\begin{aligned}\hat{V}\hat{S}^2 &= -\frac{1}{2}g\left[\sum_{pq}\hat{P}^+_p\hat{P}^-_q\right]\left[\hat{S}^2_z + \frac{1}{2}\left(\hat{S}_+\hat{S}_- + \hat{S}_-\hat{S}_+\right)\right] \\ &= \left[\hat{S}^2_z + \frac{1}{2}\left(\hat{S}_+\hat{S}_- + \hat{S}_-\hat{S}_+\right)\right]\left[-\frac{1}{2}g\sum_{pq}\hat{P}^+_p\hat{P}^-_q\right] \\ &= \hat{S}^2\hat{V}. \qquad \blacksquare\end{aligned}$$

**(5)** $\left[\hat{P}_p^+, \hat{P}_q^+\right] = 0$ **and** $\left[\hat{P}_p^-, \hat{P}_q^-\right] = 0$

$$\begin{aligned}\hat{P}_p^+ \hat{P}_q^+ &= a_{p+}^\dagger a_{p-}^\dagger \, a_{q+}^\dagger a_{q-}^\dagger \\ &= a_{p+}^\dagger a_{q+}^\dagger a_{q-}^\dagger a_{p-}^\dagger \\ &= a_{q+}^\dagger a_{q-}^\dagger a_{p+}^\dagger a_{p-}^\dagger \\ &= \hat{P}_q^+ \hat{P}_p^+ \qquad \blacksquare\end{aligned}$$

The same argument shows $\left[\hat{P}_p^-, \hat{P}_q^-\right] = 0$.

**The Hamiltonian in terms of pair operators**

$$\hat{H} = \xi \sum_{p\sigma} (p-1)\, a_{p\sigma}^\dagger a_{p\sigma} - \frac{1}{2} g \sum_{pq} a_{p+}^\dagger a_{p-}^\dagger a_{q-} a_{q+},$$

with $\hat{P}_p^+ = a_{p+}^\dagger a_{p-}^\dagger$, $\hat{P}_q^- = a_{q-} a_{q+}$ and $\xi = 1$:

$$\hat{H} = \sum_{p\sigma} (p-1)\, a_{p\sigma}^\dagger a_{p\sigma} - \frac{1}{2} g \sum_{pq} \hat{P}_p^+ \hat{P}_q^-.$$

## Task 2: Matrix elements

The four-particle Slater determinant with levels $p$ and $q$ doubly occupied is

$$\Phi_{pq}(x_1, x_2, x_3, x_4) = \begin{vmatrix} \psi_{p+}(x_1) & \psi_{p+}(x_2) & \psi_{p+}(x_3) & \psi_{p+}(x_4) \\ \psi_{p-}(x_1) & \psi_{p-}(x_2) & \psi_{p-}(x_3) & \psi_{p-}(x_4) \\ \psi_{q+}(x_1) & \psi_{q+}(x_2) & \psi_{q+}(x_3) & \psi_{q+}(x_4) \\ \psi_{q-}(x_1) & \psi_{q-}(x_2) & \psi_{q-}(x_3) & \psi_{q-}(x_4) \end{vmatrix}, \qquad |\alpha\beta\rangle = \Phi_{\alpha\beta}(x_1, x_2, x_3, x_4).$$

**Diagonal element.** Write $|\Phi\rangle = |\alpha\beta\rangle$. Then

$$\begin{aligned}\langle\Phi|\hat{H}|\Phi\rangle &= \sum_{p\sigma} (p-1) \langle\Phi| a_{p\sigma}^\dagger a_{p\sigma} |\Phi\rangle - \frac{1}{2} g \sum_{pq} \langle\Phi| a_{p+}^\dagger a_{p-}^\dagger a_{q-} a_{q+} |\Phi\rangle \\ &= \sum_{p\sigma} (p-1)\, \langle n_{p\sigma}\rangle - \frac{1}{2} g \sum_{pq} \langle\Phi|\Phi\rangle\, \delta_{pq}\, \langle n_{p+} n_{p-}\rangle \\ &= 2(\alpha-1) + 2(\beta-1) - \frac{1}{2} g \cdot 2 \\ &= \underline{2(\alpha+\beta-2) - g}.\end{aligned}$$

The one-body part counts $2(\alpha-1) + 2(\beta-1)$ because levels $\alpha$ and $\beta$ are each doubly occupied. In the pairing part, only $p = q \in \{\alpha, \beta\}$ survives, giving a factor 2.

**Pair excitation.** Let $|\Phi_i^a\rangle = a_{a+}^\dagger a_{a-}^\dagger a_{i-} a_{i+} |\Phi\rangle$ (a pair moved from level $i$ to level $a$). Then

$$\begin{aligned}
\langle\Phi|\hat{H}|\Phi_i^a\rangle &= \langle\Phi|\hat{H}\, a_{a+}^\dagger a_{a-}^\dagger a_{i-} a_{i+}|\Phi\rangle \\
&= \sum_{p\sigma}(p-1)\langle\Phi|a_{p\sigma}^\dagger a_{p\sigma}\, a_{a+}^\dagger a_{a-}^\dagger a_{i-} a_{i+}|\Phi\rangle \\
&\quad - \frac{1}{2}g\sum_{pq}\langle\Phi|a_{p+}^\dagger a_{p-}^\dagger a_{q-} a_{q+}\, a_{a+}^\dagger a_{a-}^\dagger a_{i-} a_{i+}|\Phi\rangle \\
&= \sum_{p\sigma}(p-1)\langle\Phi|\Phi\rangle\cdot 0 - \frac{1}{2}g\sum_{pq}\langle\Phi|\Phi\rangle\,\delta_{pi}\delta_{qa} \\
&= 0 - \frac{1}{2}g\cdot 1 \\
&= \underline{-\frac{1}{2}g.}
\end{aligned}$$

The one-body term vanishes because $\hat{H}_0$ is diagonal in the determinant basis and $|\Phi_i^a\rangle \neq |\Phi\rangle$.

**Double pair excitation.**

$$\langle\Phi|\hat{H}|\Phi_{ij}^{ab}\rangle = 0.$$

**Matrix elements.** From Task 2, with $|\alpha\beta\rangle$ the determinant in which levels $\alpha$ and $\beta$ are doubly occupied,

$$\begin{aligned}
\langle\Phi|\hat{H}|\Phi\rangle &= 2(\alpha+\beta-2) - g, \\
\langle\Phi|\hat{H}|\Phi_i^a\rangle &= -\tfrac{1}{2}g, \\
\langle\Phi|\hat{H}|\Phi_{ij}^{ab}\rangle &= 0.
\end{aligned}$$

Two determinants coupled by a single pair excitation therefore have matrix element $-\frac{1}{2}g$, and determinants that differ by both pairs have matrix element 0.

**Full basis (exact calculation).** In the basis $\{|12\rangle, |13\rangle, |14\rangle, |23\rangle, |24\rangle, |34\rangle\}$,

$$\hat{H} = \begin{array}{c|cccccc}
 & |12\rangle & |13\rangle & |14\rangle & |23\rangle & |24\rangle & |34\rangle \\ \hline
|12\rangle & 2-g & -\frac{1}{2}g & -\frac{1}{2}g & -\frac{1}{2}g & -\frac{1}{2}g & 0 \\
|13\rangle & -\frac{1}{2}g & 4-g & -\frac{1}{2}g & -\frac{1}{2}g & 0 & -\frac{1}{2}g \\
|14\rangle & -\frac{1}{2}g & -\frac{1}{2}g & 6-g & 0 & -\frac{1}{2}g & -\frac{1}{2}g \\
|23\rangle & -\frac{1}{2}g & -\frac{1}{2}g & 0 & 6-g & -\frac{1}{2}g & -\frac{1}{2}g \\
|24\rangle & -\frac{1}{2}g & 0 & -\frac{1}{2}g & -\frac{1}{2}g & 8-g & -\frac{1}{2}g \\
|34\rangle & 0 & -\frac{1}{2}g & -\frac{1}{2}g & -\frac{1}{2}g & -\frac{1}{2}g & 10-g
\end{array}$$

The exact ground-state energy decreases as the coupling strength $g$ increases over the interval $g \in [-1, 1]$ as seen in Figure 1. The energy depends nonlinearly on $g$, indicating that the relationship between the interaction strength and the ground-state energy is not simply linear. This behavior arises because the interaction affects the ground-state wavefunction, changing the contributions of different configurations as $g$ varies.

## Task 3: Truncated Hamiltonian matrix

**Truncation to at most 2p–2h excitations.** Taking $|12\rangle$ as the reference and keeping only the reference and the four single-pair excitations (dropping $|34\rangle$, the 4p–4h state), we get the

$5 \times 5$ matrix

$$\hat{H} = \begin{array}{c|ccccc} & |12\rangle & |13\rangle & |14\rangle & |23\rangle & |24\rangle \\ \hline |12\rangle & 2-g & -\frac{1}{2}g & -\frac{1}{2}g & -\frac{1}{2}g & -\frac{1}{2}g \\ |13\rangle & -\frac{1}{2}g & 4-g & -\frac{1}{2}g & -\frac{1}{2}g & 0 \\ |14\rangle & -\frac{1}{2}g & -\frac{1}{2}g & 6-g & 0 & -\frac{1}{2}g \\ |23\rangle & -\frac{1}{2}g & -\frac{1}{2}g & 0 & 6-g & -\frac{1}{2}g \\ |24\rangle & -\frac{1}{2}g & 0 & -\frac{1}{2}g & -\frac{1}{2}g & 8-g \end{array}$$

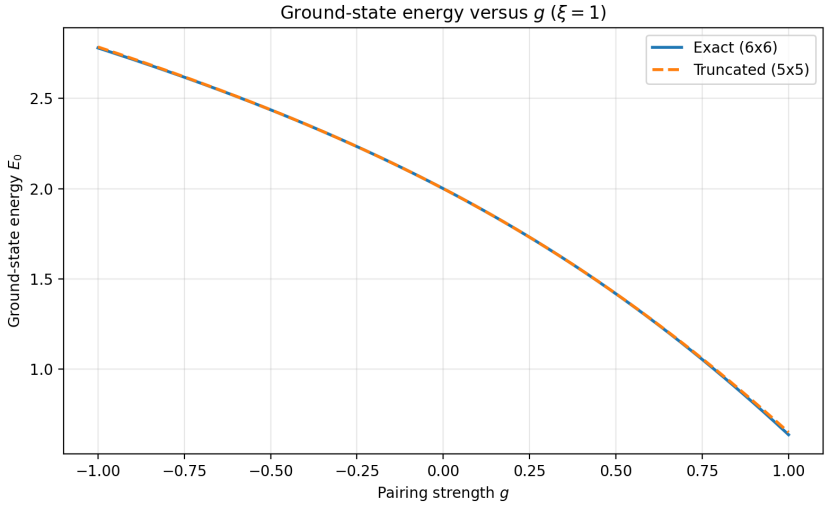



Figure 1: Exact and truncated FCI ground-state energies as functions of the coupling strength $g \in [-1, 1]$. The truncated ground-state energy is slightly higher than the exact ground-state energy (GitHub).

The ground-state energy obtained by diagonalizing the truncated Hamiltonian matrix is slightly higher than the exact ground-state energy throughout the considered range of $g$ (Figure 1). This is expected because the truncated calculation uses a restricted configuration space, whereas the exact calculation includes the full configuration space. By the variational principle, restricting the configuration space cannot produce a ground-state energy lower than the exact ground-state energy, provided both calculations use the same Hamiltonian and the truncation is treated variationally.

Despite this small difference in energy, the truncated calculation reproduces the overall behavior of the exact ground-state energy as a function of $g$. This suggests that the truncated configuration space captures the dominant contributions to the ground state, although some correlation effects are missing.

# Part 2: Hartree–Fock theory

## Task 4: Hartree–Fock theory

### (a) Normal-ordered Hamiltonian

The Hamiltonian is $\hat{H} = \hat{H}_0 + \hat{V}$ with

$$\hat{H}_0 = \sum_{p\sigma} (p-1)\, a^\dagger_{p\sigma} a_{p\sigma}, \qquad \hat{V} = -\frac{1}{2} g \sum_{pq} a^\dagger_{p+} a^\dagger_{p-} a_{q-} a_{q+}.$$

Write $p < F \equiv i$ for the occupied levels $(1, 2)$ and $p > F \equiv a$ for the unoccupied levels $(3, 4)$. In the reference state $|12\rangle$,

$$\langle a^\dagger_{i\sigma} a_{i\sigma} \rangle = 1, \qquad \langle a^\dagger_{a\sigma} a_{a\sigma} \rangle = 0.$$

We reorder the creation and annihilation operators so that they are normal ordered with respect to the ground state $|12\rangle$ instead of the vacuum $|0\rangle$. Since the levels $p < F$ are occupied, $a^\dagger_{i\sigma}$ annihilates $|12\rangle$, and we want all operators that annihilate the reference state to the right:

$$a_{i\sigma} a^\dagger_{i\sigma} |12\rangle = 0 \quad (p < F), \qquad \mathcal{N}\Big(a^\dagger_{i\sigma} a_{i\sigma}\Big) = -a_{i\sigma} a^\dagger_{i\sigma}, \qquad \mathcal{N}\Big(a^\dagger_{a\sigma} a_{a\sigma}\Big) = a^\dagger_{a\sigma} a_{a\sigma}.$$

**One-body part.**

$$\begin{aligned}
\hat{H}_0 &= \sum_{i\sigma} (i-1)\Big(1 - a_{i\sigma} a^\dagger_{i\sigma}\Big) + \sum_{a\sigma} (a-1)\, a^\dagger_{a\sigma} a_{a\sigma} \\
&= \sum_{i\sigma} (i-1) + \sum_{p\sigma} (p-1)\, \mathcal{N}\Big(a^\dagger_{p\sigma} a_{p\sigma}\Big) \\
&= \sum_{i} 2(i-1) + \sum_{p\sigma} (p-1)\, \mathcal{N}\Big(a^\dagger_{p\sigma} a_{p\sigma}\Big) \\
&= 2(1-1) + 2(2-1) + \sum_{p\sigma} (p-1)\, \mathcal{N}\Big(a^\dagger_{p\sigma} a_{p\sigma}\Big) \\
&= 2 + \sum_{p\sigma} (p-1)\, \mathcal{N}\Big(a^\dagger_{p\sigma} a_{p\sigma}\Big).
\end{aligned}$$

**Pairing part.** Using Wick's theorem relative to $|12\rangle$,

$$\begin{aligned}
a^\dagger_{p+} a^\dagger_{p-} a_{q-} a_{q+} = {} & \mathcal{N}\Big(a^\dagger_{p+} a^\dagger_{p-} a_{q-} a_{q+}\Big) \\
& + \delta_{pq}\, \delta_{p,q<F}\, \mathcal{N}\Big(a^\dagger_{p+} a_{q+}\Big) \qquad \text{(contraction } a^\dagger_{p-}\, a_{q-}) \\
& + 0 \qquad (a^\dagger_{p-}\, a_{q+}) \\
& + 0 \qquad (a^\dagger_{p+}\, a_{q-}) \\
& + \delta_{pq}\, \delta_{p,q<F}\, \mathcal{N}\Big(a^\dagger_{p-} a_{q-}\Big) \qquad (a^\dagger_{p+}\, a_{q+}) \\
& + \delta_{pq}\, \delta_{p,q<F} \qquad \text{(both } a^\dagger_{p+} a_{q+} \text{ and } a^\dagger_{p-} a_{q-}) \\
& + 0 \qquad \text{(both cross contractions).}
\end{aligned}$$

The cross contractions vanish because the spins differ. Multiplying by $-\frac{1}{2} g$ and summing over $p, q$,

$$\begin{aligned}
\hat{V} &= -\frac{1}{2} g \Big[ \sum_i 1 + \mathcal{N}\Big(a^\dagger_{i+} a_{i+}\Big) + \mathcal{N}\Big(a^\dagger_{i-} a_{i-}\Big) + \sum_{pq} \mathcal{N}\Big(a^\dagger_{p+} a^\dagger_{p-} a_{q-} a_{q+}\Big) \Big] \\
&= -g - \frac{1}{2} g \sum_{i\sigma} \mathcal{N}\Big(a^\dagger_{i\sigma} a_{i\sigma}\Big) - \frac{1}{2} g \sum_{pq} \mathcal{N}\Big(a^\dagger_{p+} a^\dagger_{p-} a_{q-} a_{q+}\Big).
\end{aligned}$$

**Result.**

$$\begin{aligned}
\hat{H} &= E_{\text{ref}} + \hat{F}_N + \hat{V}_N, \\
E_{\text{ref}} &= 2 - g, \\
\hat{F}_N &= \sum_{p\sigma} (p-1)\,\mathcal{N}\Big(a^\dagger_{p\sigma} a_{p\sigma}\Big) - \frac{1}{2} g \sum_{i\sigma} \mathcal{N}\Big(a^\dagger_{i\sigma} a_{i\sigma}\Big), \\
\hat{V}_N &= -\frac{1}{2} g \sum_{pq} \mathcal{N}\Big(a^\dagger_{p+} a^\dagger_{p-} a_{q-} a_{q+}\Big).
\end{aligned}$$

### (b) The Hartree–Fock equations

**Energy functional.** Expand the new single-particle states in the original basis, $|p\rangle = \sum_\alpha C_{p\alpha} |\alpha\rangle$. The energy of the Slater determinant built from the occupied states $p$ is

$$E = \sum_p \langle p|h|p\rangle + \frac{1}{2} \sum_{pq} \langle pq|V|pq\rangle_{AS}.$$

Inserting the original basis,

$$E = \sum_p \sum_{\alpha\gamma} C^*_{p\alpha} C_{p\gamma} \langle \alpha|h|\gamma\rangle + \frac{1}{2} \sum_{pq} \sum_{\alpha\beta\gamma\delta} C^*_{p\alpha} C^*_{q\beta} C_{p\gamma} C_{q\delta}\, \langle \alpha\beta|V|\gamma\delta\rangle_{AS}.$$

**Constraint.** The new basis must stay orthonormal,

$$\sum_\alpha C_{p\alpha} C^*_{p\alpha} = 1,$$

which we include with Lagrange multipliers $\epsilon^{HF}_p$:

$$\mathcal{L} = E - \sum_p \epsilon^{HF}_p \Big( \sum_\alpha C^*_{p\alpha} C_{p\alpha} - 1 \Big).$$

**Minimizing with respect to $C^*_{p\alpha}$.** Since $C^*_{p\alpha}$ and $C_{p\alpha}$ are treated as independent,

$$\begin{aligned}
&\frac{\partial \mathcal{L}}{\partial C^*_{p\alpha}} = \sum_\gamma C_{p\gamma} \langle \alpha|h|\gamma\rangle + \sum_q \sum_{\beta\gamma\delta} C^*_{q\beta} C_{p\gamma} C_{q\delta} \langle \alpha\beta|V|\gamma\delta\rangle_{AS} - \epsilon^{HF}_p C_{p\alpha} = 0 \\
&\Rightarrow \quad \sum_\gamma \Big[ \langle \alpha|h|\gamma\rangle + \sum_q \sum_{\beta\delta} C^*_{q\beta} C_{q\delta} \langle \alpha\beta|V|\gamma\delta\rangle_{AS} \Big] C_{p\gamma} = \epsilon^{HF}_p C_{p\alpha}.
\end{aligned}$$

With

$$h^{HF}_{\alpha\gamma} = \langle \alpha|h|\gamma\rangle + \sum_q \sum_{\beta\delta} C^*_{q\beta} C_{q\delta} \langle \alpha\beta|V|\gamma\delta\rangle_{AS}$$

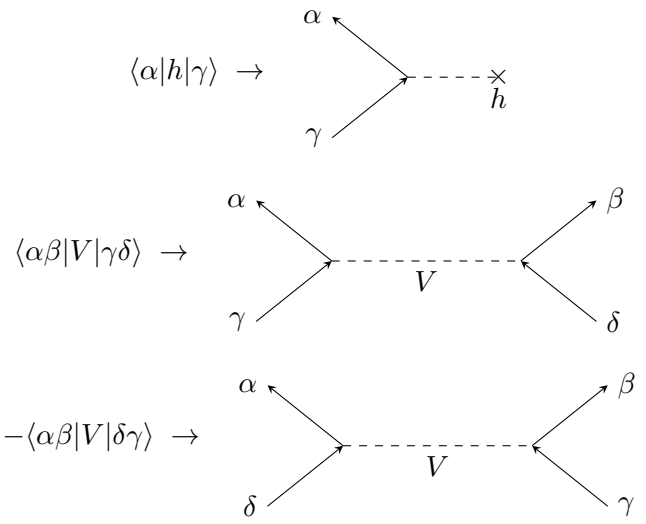
this becomes the Hartree–Fock equation

$$\boxed{\sum_\gamma h^{HF}_{\alpha\gamma} C_{p\gamma} = \epsilon^{HF}_p C_{p\alpha}}$$

**Meaning of the terms.**

- $\langle \alpha|h|\gamma\rangle$: the one-body part. It describes how each particle behaves on its own.
- $\langle \alpha\beta|V|\gamma\delta\rangle$: the two-body (direct) term. It describes the potential felt by a particle, created by the other particles.
- $-\langle \alpha\beta|V|\delta\gamma\rangle$: the two-body (exchange) term, a consequence of the antisymmetry of fermions.

**Diagrammatic representation.**




**The Hartree–Fock operator in second quantization.**

$$\hat{F} = \sum_{\alpha\gamma} h^{HF}_{\alpha\gamma}\, a^\dagger_\alpha a_\gamma.$$

**(c) Setting up and solving the equations**

For $\hat{H}_0$ the matrix elements are

$$\langle\alpha|\hat{H}_0|\beta\rangle = \xi \sum_{p\sigma}(p-1)\langle 0|a_\alpha\, a^\dagger_{p\sigma} a_{p\sigma}\, a^\dagger_\beta|0\rangle = \xi\,(p_\alpha - 1)\,\delta_{\alpha\beta},$$

with $\xi = 1$. We can then write the condition as the matrix equation

$$h^{HF}C = \epsilon^{HF}C \qquad \text{i.e.} \qquad \sum_\gamma h^{HF}_{\alpha\gamma} C_{p\gamma} = \epsilon^{HF}_p C_{p\alpha}.$$

Since $h^{HF}$ itself depends on $C$, we start with the guess $C^{(0)} = I$, use it to build $h^{HF}$, find $C^{(1)}$ by solving the matrix equation, and repeat until $\epsilon^{HF}$ no longer changes. We now do one iteration by hand, with $C^{(0)} = I$.

**One-body part of $h^{HF}$.**

$$\langle\alpha|\hat{H}_0|\gamma\rangle = (p_\alpha - 1)\,\delta_{\alpha\gamma} \quad \Rightarrow \quad h^{HF}_{\text{one-body}} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

**Two-body part of $h^{HF}$.** With $C^{(0)} = I$ the sums collapse,

$$\sum_p \sum_{\beta\delta} C^*_{p\beta} C_{p\delta} \langle\alpha\beta|V|\gamma\delta\rangle_{AS} = \sum_p \sum_{\beta\delta} \delta_{p\beta}\,\delta_{p\delta}\langle\alpha\beta|V|\gamma\delta\rangle_{AS} = \sum_p \langle\alpha p|V|\gamma p\rangle_{AS} = \sum_i \langle\alpha i|V|\gamma i\rangle_{AS},$$

where $p$ has been replaced by $i$ since $p < F$.

The interaction only acts on a pair of particles on the same level with opposite spin, and $i$ is the same on both sides. Hence $\alpha$ and $\gamma$ must both be the spin partner of $i$, so $\alpha = \gamma$ and all off-diagonal elements are 0. Since the partner $i$ must be occupied, $\alpha$ must lie on an occupied

level $(1,2)$. For such $\alpha$ exactly one $i$ contributes. The value is read off from the one-body part of the normal-ordered $\hat{V}$, $-\frac{1}{2}g\sum_{i\sigma}\mathcal{N}(a^\dagger_{i\sigma}a_{i\sigma})$, so

$$h^{HF}_{\alpha\alpha}\Big|_{\text{two-body}} = \sum_i \langle \alpha i|V|\alpha i\rangle_{AS} = \begin{cases} -\frac{1}{2}g & \alpha \in \text{occupied}, \\ 0 & \text{otherwise}, \end{cases}$$

$$h^{HF}_{\text{two-body}} = \begin{pmatrix} -\frac{1}{2}g & 0 & 0 & 0 \\ 0 & -\frac{1}{2}g & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad h^{HF(0)} = \begin{pmatrix} -\frac{1}{2}g & 0 & 0 & 0 \\ 0 & 1-\frac{1}{2}g & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

**Solving.**

$$h^{HF(0)}C^{(1)} = \epsilon^{HF(1)}C^{(1)}.$$

Since $h^{HF(0)}$ is already diagonal,

$$C^{(1)} = I = C^{(0)}, \qquad \epsilon^{HF(1)} = \begin{pmatrix} -\frac{1}{2}g \\ 1-\frac{1}{2}g \\ 2 \\ 3 \end{pmatrix}.$$

Since $C$ is the same in every iteration, $\epsilon^{HF}$ and $h^{HF}$ are also the same, so one iteration was all that was needed. The Hartree–Fock basis equals the original basis, and the Hartree–Fock ground state is $|12\rangle$.

**Hartree–Fock energy.** Here $p, q$ run over the occupied spin-orbitals,

$$\begin{aligned} E_{HF} &= \sum_p \epsilon^{HF}_p - \frac{1}{2}\sum_{pq}\langle pq|\hat{V}|pq\rangle_{AS} \\ &= 2 - 2g - \Big(\frac{1}{2}\cdot 4\cdot\big(-\frac{g}{2}\big)\Big) \\ &= 2 - g. \end{aligned}$$

This script at Github shows that the Hartree–Fock equations converge in a single iteration for all tested values of $g \in [-1, 1]$. The initial guess $C = I$ is therefore already self-consistent, the occupied orbitals are unchanged when the Hartree–Fock matrix is diagonalized.

Figure 2 shows that the numerical Hartree–Fock energies agree exactly with the analytical result $E_{\text{HF}} = 2 - g$ (for $\xi = 1$). The Hartree–Fock energy decreases linearly with $g$, whereas the exact ground-state energy depends nonlinearly on $g$. This illustrates the limitation of the Hartree–Fock approximation, a single Slater determinant captures the overall decrease in energy but not the correlations included in the exact calculation.

## Part 3: Matrix product operators

### Task 5: Setting up the MPO

$$\hat{H} = \sum_{p=1}^{L}\varepsilon_p\hat{n}_p - \frac{g}{2}\sum_{pq}\hat{P}^+_p\hat{P}^-_q, \qquad \varepsilon_p = \xi\,(p-1).$$

$\varepsilon_p$ is the energy of site $p$ and $\hat{n}_p$ checks whether site $p$ is occupied. $\sum_{p=1}^{L}\varepsilon_p\hat{n}_p$ is the sum of the single-particle energies of all states occupied in $|n_1 n_2 \ldots n_L\rangle$. $\hat{P}^+_p\hat{P}^-_q$ removes the pair at

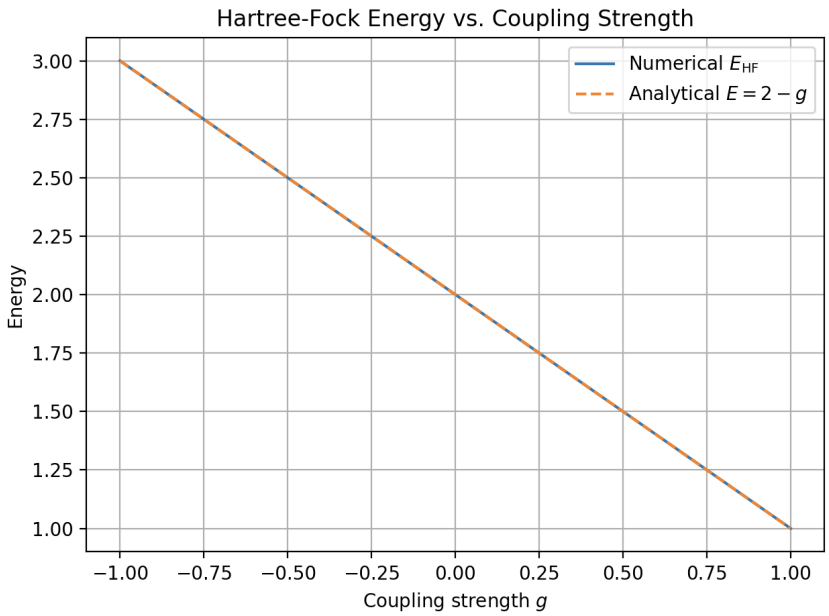



Figure 2: Hartree–Fock ground-state energy as a function of the coupling strength $g$. The numerical results agree with the analytical expression $E_{\mathrm{HF}} = 2 - g$ (GitHub).

position $q$ and creates a pair at position $p$. $g$ is the pairing interaction strength. $-\frac{g}{2}\sum_{pq}\hat{P}_p^+\hat{P}_q^-$ sums over all pairs, where a pair at site $q$ can be annihilated and a pair at site $p$ can be created, each contributing with $-\frac{g}{2}$.

Local basis $\{|0\rangle, |1\rangle\}$ with $|0\rangle = \begin{pmatrix}1\\0\end{pmatrix}$, $|1\rangle = \begin{pmatrix}0\\1\end{pmatrix}$:

$$\left.\begin{aligned}\hat{P}_p^+|0\rangle_p &= |1\rangle_p\\ \hat{P}_p^+|1\rangle_p &= \vec{0}_p\end{aligned}\right\}\ \hat{P}_p^+ = \begin{pmatrix}0 & 0\\ 1 & 0\end{pmatrix}, \qquad \left.\begin{aligned}\hat{P}_p^-|1\rangle_p &= |0\rangle_p\\ \hat{P}_p^-|0\rangle_p &= \vec{0}_p\end{aligned}\right\}\ \hat{P}_p^- = \begin{pmatrix}0 & 1\\ 0 & 0\end{pmatrix}$$

$$\hat{n}_p = \hat{n}_{p+} + \hat{n}_{p-}, \qquad \left.\begin{aligned}\hat{n}_p|0\rangle_p &= 0\,|0\rangle_p\\ \hat{n}_p|1\rangle_p &= 2\,|1\rangle_p\end{aligned}\right\}\ \hat{n}_p = \begin{pmatrix}0 & 0\\ 0 & 2\end{pmatrix}$$

**States of the automaton**

1. Every site so far did nothing.
2. $\hat{P}^+$ applied to this site or a previous one. Some site to the right must apply $\hat{P}^-$ (start term).
3. $\hat{P}^-$ applied to this site or a previous one. Some site to the right must apply $\hat{P}^+$ (start term).
4. Term is done, do nothing from here.

$D = 4$ since between two neighbouring sites the left side only needs one of four possible "messages". This is independent of $L$ since the messages never record which site started a term.

**The MPO matrices**

$W^{[p]}$ is a $4 \times 4$ matrix where element $(\alpha, \beta)$ is the operator applied when going from state of automaton $\alpha$ to state of automaton $\beta$.

$$W^{[p]} = \begin{pmatrix} I & \hat{P}_p^+ & \hat{P}_p^- & \varepsilon_p \hat{n}_p - \frac{g}{2}\hat{P}_p^+\hat{P}_p^- \\ 0 & I & 0 & -\frac{g}{2}\hat{P}_p^- \\ 0 & 0 & I & -\frac{g}{2}\hat{P}_p^+ \\ 0 & 0 & 0 & I \end{pmatrix}$$

With $X_p \equiv \varepsilon_p \hat{n}_p - \frac{g}{2}\hat{P}_p^+\hat{P}_p^-$, the first and last sites use the first row and last column:

$$W^{[1]} = \begin{pmatrix} I & \hat{P}_1^+ & \hat{P}_1^- & X_1 \end{pmatrix}, \qquad W^{[4]} = \begin{pmatrix} X_4 \\ -\frac{g}{2}\hat{P}_4^- \\ -\frac{g}{2}\hat{P}_4^+ \\ I \end{pmatrix}.$$

**Two sites.**

$$W^{[1]}W^{[2]} = \begin{pmatrix} I & \hat{P}_1^+ & \hat{P}_1^- & X_1 \end{pmatrix} \begin{pmatrix} I & \hat{P}_2^+ & \hat{P}_2^- & X_2 \\ 0 & I & 0 & -\frac{g}{2}\hat{P}_2^- \\ 0 & 0 & I & -\frac{g}{2}\hat{P}_2^+ \\ 0 & 0 & 0 & I \end{pmatrix}$$

$$= \Big( I,\ \hat{P}_1^+ + \hat{P}_2^+,\ \hat{P}_1^- + \hat{P}_2^-,\ X_2 - \tfrac{g}{2}\hat{P}_1^+\hat{P}_2^- - \tfrac{g}{2}\hat{P}_1^-\hat{P}_2^+ + X_1 \Big).$$

**Three sites.**

$$W^{[1]}W^{[2]}W^{[3]} = \Big( I,\ \hat{P}_1^+ + \hat{P}_2^+ + \hat{P}_3^+,\ \hat{P}_1^- + \hat{P}_2^- + \hat{P}_3^-,\ X_3 - \tfrac{g}{2}\hat{P}_1^+\hat{P}_3^- - \tfrac{g}{2}\hat{P}_2^+\hat{P}_3^- - \tfrac{g}{2}\hat{P}_1^-\hat{P}_3^+ - \tfrac{g}{2}\hat{P}_2^-\hat{P}_3^+ + X_2 - \tfrac{g}{2}\hat{P}_1^+\hat{P}_2^- - \tfrac{g}{2}\hat{P}_1^-\hat{P}_2^+ + X_1 \Big).$$

**Four sites.**

$$\begin{aligned} W^{[1]}W^{[2]}W^{[3]}W^{[4]} &= X_4 + X_3 + X_2 + X_1 \\ &\quad - \frac{g}{2}\Big[\hat{P}_1^+\hat{P}_4^- + \hat{P}_2^+\hat{P}_4^- + \hat{P}_3^+\hat{P}_4^- + \hat{P}_4^+\hat{P}_1^- + \hat{P}_4^+\hat{P}_2^- + \hat{P}_4^+\hat{P}_3^- \\ &\qquad + \hat{P}_1^+\hat{P}_3^- + \hat{P}_2^+\hat{P}_3^- + \hat{P}_3^+\hat{P}_1^- + \hat{P}_3^+\hat{P}_2^- + \hat{P}_1^+\hat{P}_2^- + \hat{P}_2^+\hat{P}_1^-\Big] \\ &= \sum_{p=1}^{4} \varepsilon_p \hat{n}_p - \frac{g}{2}\sum_{pq} \hat{P}_p^+\hat{P}_q^- = \hat{H}. \end{aligned}$$

**Constructing the Hamiltonian**

1. $\varepsilon_p \hat{n}_p$ gives $2\varepsilon_p$ if $p$ is occupied (a doubly occupied level has $\hat{n}_p = 2$).
2. $-\frac{g}{2}\hat{P}_p^+\hat{P}_q^-$ gives $-\frac{g}{2}$ if $p = q$ and $p$ is occupied.
3. $-\frac{g}{2}\hat{P}_p^+\hat{P}_q^-$ $(p \neq q)$ removes a pair at $q$ and adds it to $p$. This gives 0 if $q$ is unoccupied or $p$ is occupied.

$$\hat{H}|1100\rangle = [2(\varepsilon_1+\varepsilon_2)-g]|1100\rangle - \frac{g}{2}\big[|1010\rangle+|0110\rangle+|1001\rangle+|0101\rangle\big] + 0\,|0011\rangle$$

$$\hat{H}|1010\rangle = [2(\varepsilon_1+\varepsilon_3)-g]|1010\rangle - \frac{g}{2}\big[|1100\rangle+|0110\rangle+|1001\rangle+|0011\rangle\big] + 0\,|0101\rangle$$

$$\hat{H}|0110\rangle = [2(\varepsilon_2+\varepsilon_3)-g]|0110\rangle - \frac{g}{2}\big[|1100\rangle+|1010\rangle+|0101\rangle+|0011\rangle\big] + 0\,|1001\rangle$$

$$\hat{H}|1001\rangle = [2(\varepsilon_1+\varepsilon_4)-g]|1001\rangle - \frac{g}{2}\big[|1100\rangle+|1010\rangle+|0101\rangle+|0011\rangle\big] + 0\,|0110\rangle$$

$$\hat{H}|0101\rangle = [2(\varepsilon_2+\varepsilon_4)-g]|0101\rangle - \frac{g}{2}\big[|1100\rangle+|0110\rangle+|1001\rangle+|0011\rangle\big] + 0\,|1010\rangle$$

$$\hat{H}|0011\rangle = [2(\varepsilon_3+\varepsilon_4)-g]|0011\rangle - \frac{g}{2}\big[|1010\rangle+|0110\rangle+|1001\rangle+|0101\rangle\big] + 0\,|1100\rangle$$

In the basis $|1100\rangle, |1010\rangle, |0110\rangle, |1001\rangle, |0101\rangle, |0011\rangle$:

$$\hat{H} = \begin{pmatrix} 2(\varepsilon_1+\varepsilon_2)-g & -\frac{g}{2} & -\frac{g}{2} & -\frac{g}{2} & -\frac{g}{2} & 0 \\ -\frac{g}{2} & 2(\varepsilon_1+\varepsilon_3)-g & -\frac{g}{2} & -\frac{g}{2} & 0 & -\frac{g}{2} \\ -\frac{g}{2} & -\frac{g}{2} & 2(\varepsilon_2+\varepsilon_3)-g & 0 & -\frac{g}{2} & -\frac{g}{2} \\ -\frac{g}{2} & -\frac{g}{2} & 0 & 2(\varepsilon_1+\varepsilon_4)-g & -\frac{g}{2} & -\frac{g}{2} \\ -\frac{g}{2} & 0 & -\frac{g}{2} & -\frac{g}{2} & 2(\varepsilon_2+\varepsilon_4)-g & -\frac{g}{2} \\ 0 & -\frac{g}{2} & -\frac{g}{2} & -\frac{g}{2} & -\frac{g}{2} & 2(\varepsilon_3+\varepsilon_4)-g \end{pmatrix}$$

With $\xi = 1$ this is the same $\hat{H}$ as in Task 2 (up to the order of $|0110\rangle$ and $|1001\rangle$).

**Increasing the local dimension**

If we increase $d$ from $d = 2$ to $d = 4$, $D$ will still be $D = 4$. The size of $W$ stays the same, $D \times D$, but the size of $I, \hat{P}_p^+, \hat{P}_p^-, \hat{n}_p$ must increase from $2 \times 2$ to $4 \times 4$.

## Task 6: MPO for fermions

In MPOs, operators on different sites must commute. For fermions, the Hamiltonian often has operators that anticommute, like $\{a_p, a_q\}$ and $\{a_p^\dagger, a_q^\dagger\}$. To fix this, we add Jordan–Wigner transformations on the sites in between, which make the operators commute between sites. These chains of Jordan–Wigner transformations are called a *parity string*.

Luckily our Hamiltonian only has operators that commute, as shown in Task 1, so we avoid this problem. It does not matter whether $d = 2$ or $d = 4$: our Hamiltonian does not allow single-fermion excitations and the operators commute.

If we do allow single-particle excitations (for example $a_{p\sigma}^\dagger a_{q\sigma}$) we must make sure the operators commute between sites. This is done by adding a Jordan–Wigner transformation that creates a parity string between the operators. These transformations appear in the waiting entries of the MPO. For our $W^{[p]}$ it would look something like

$$W^{[p]} = \begin{pmatrix} I & \hat{P}_p^+ & \hat{P}_p^- & \varepsilon_p\hat{n}_p - (g/2)\hat{P}_p^+\hat{P}_p^- \\ 0 & F & 0 & -(g/2)\hat{P}_p^- \\ 0 & 0 & F & -(g/2)\hat{P}_p^+ \\ 0 & 0 & 0 & I \end{pmatrix}.$$

Here $F$ is the parity matrix, and in this case we assume $\{\hat{P}_p^+, \hat{P}_q^+\}$ and $\{\hat{P}_p^-, \hat{P}_q^-\}$ (i.e. that these operators anticommute).

## Task 7: Matrix product states by hand, the degenerate limit

### (i) Ground state for $\xi = 0$

With $\xi = 0$ the Hamiltonian is

$$\hat{H} = -\frac{g}{2} \sum_{pq} \hat{P}_p^+ \hat{P}_q^- .$$

In the basis $|1100\rangle, |1010\rangle, |0110\rangle, |1001\rangle, |0101\rangle, |0011\rangle$,

$$\hat{H} = -\frac{1}{2}\, g \begin{pmatrix} 2 & 1 & 1 & 1 & 1 & 0 \\ 1 & 2 & 1 & 1 & 0 & 1 \\ 1 & 1 & 2 & 0 & 1 & 1 \\ 1 & 1 & 0 & 2 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 1 & 1 & 2 \end{pmatrix}.$$

We want $\hat{H}|\psi\rangle = E|\psi\rangle$ where $E$ is smallest, since this gives the ground state. $\langle\psi|\psi\rangle$ must also be 1. For this, $|\psi\rangle$ is the superposition of all states:

$$|\psi\rangle = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \qquad \hat{H}|\psi\rangle = -3g\,|\psi\rangle, \qquad E_0 = -3g.$$

We can also verify numerically that this is the ground state. Using the numerical results from GitHub, we verify that the eigenvector that is superposition of all states has the lowest eigenvalue.

For $\xi = 1$ the one-body energies dominate for $g \in [-1, 1]$, and the ground state changes smoothly with $g$. It remains mostly $|1100\rangle$ for all $g \in [-1, 1]$, which is why $|1100\rangle$ is a good Hartree–Fock reference for both signs of $g$. For $\xi = 0$ the Hamiltonian is proportional to $g$, so the eigenvectors do not depend on $g$ and only the ordering of the eigenvalues changes with its sign. For $g > 0$ the equal superposition has the lowest energy $-3g$, while for $g < 0$ the same state has the highest energy $-3g > 0$. So the equal superposition is only the ground state for $g > 0$. For $g < 0$ the ground state is instead a different, degenerate combination of the states, and at $g = 0$ all states are degenerate.

### (ii) Coefficient matrices and the MPS in left-canonical form

Write the state as $|n_1 n_2 n_3 n_4\rangle$ and let $X = 1/\sqrt{6}$. The coefficient matrices for the three cuts are

$$C^{(1|234)} = \begin{array}{c} \\ |0\rangle \\ |1\rangle \end{array} \begin{array}{c} \begin{array}{cccccccc} |000\rangle & |100\rangle & |010\rangle & |110\rangle & |001\rangle & |101\rangle & |011\rangle & |111\rangle \end{array} \\ \begin{pmatrix} 0 & 0 & 0 & X & 0 & X & X & 0 \\ 0 & X & X & 0 & X & 0 & 0 & 0 \end{pmatrix} \end{array},$$

$$C^{(12|34)} = \begin{array}{c} \\ |00\rangle \\ |10\rangle \\ |01\rangle \\ |11\rangle \end{array} \begin{array}{c} \begin{array}{cccc} |00\rangle & |10\rangle & |01\rangle & |11\rangle \end{array} \\ \begin{pmatrix} 0 & 0 & 0 & X \\ 0 & X & X & 0 \\ 0 & X & X & 0 \\ X & 0 & 0 & 0 \end{pmatrix} \end{array},$$

$$C^{(123|4)} = \begin{array}{c} \\ |000\rangle \\ |100\rangle \\ |010\rangle \\ |110\rangle \\ |001\rangle \\ |101\rangle \\ |011\rangle \\ |111\rangle \end{array} \begin{array}{c} \begin{array}{cc} |0\rangle & |1\rangle \end{array} \\ \begin{pmatrix} 0 & 0 \\ 0 & X \\ 0 & X \\ X & 0 \\ 0 & X \\ X & 0 \\ X & 0 \\ 0 & 0 \end{pmatrix} \end{array}.$$

The SVD calculations below were not done by hand, it was done numerically using GitHub.

**First cut.**

$$\begin{aligned} C^{(1|234)} = C_1 &= U_1 \Sigma_1 V_1^\dagger, \\ C_1 &= \frac{1}{\sqrt{6}} \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \end{pmatrix}, \\ U_1 &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \Sigma_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \\ V_1^\dagger &= \frac{1}{\sqrt{3}} \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \end{pmatrix}, \\ \Sigma_1 V_1^\dagger &= \frac{1}{\sqrt{6}} \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \end{pmatrix} = C_1. \end{aligned}$$

**Second cut.** Reshape $\Sigma_1 V_1^\dagger$ into a $4 \times 4$ matrix:

$$\begin{aligned} C_2 &= \frac{1}{\sqrt{6}} \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} = U_2 \Sigma_2 V_2^\dagger, \\ \Sigma_2 &= \frac{1}{\sqrt{6}} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \\ U_2 &= \begin{pmatrix} 0 & 0 & -1 \\ -\frac{1}{\sqrt{2}} & 0 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & -1 & 0 \end{pmatrix}, \qquad V_2^\dagger = \begin{pmatrix} 0 & -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \\ \Sigma_2 V_2^\dagger &= \begin{pmatrix} 0 & -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} & 0 \\ -\frac{1}{\sqrt{6}} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{\sqrt{6}} \end{pmatrix}. \end{aligned}$$

**Third cut.** Reshape $\Sigma_2 V_2^\dagger$ into a $6 \times 2$ matrix:

$$C_3 = \begin{pmatrix} 0 & -\frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{3}} & 0 \\ -\frac{1}{\sqrt{6}} & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & -\frac{1}{\sqrt{6}} \end{pmatrix} = U_3 \Sigma_3 V_3^\dagger,$$

$$\Sigma_3 = \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} \end{pmatrix}, \qquad V_3^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

$$U_3 = \begin{pmatrix} 0 & -\sqrt{\frac{2}{3}} \\ -\sqrt{\frac{2}{3}} & 0 \\ -\frac{1}{\sqrt{3}} & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & -\frac{1}{\sqrt{3}} \end{pmatrix}.$$

The last tensor is the remainder

$$U_4 = \Sigma_3 V_3^\dagger = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

**The tensors $A^{[i]s_i}$.**

$$A^{[1]0} = \begin{pmatrix} 1 & 0 \end{pmatrix}, \qquad A^{[1]1} = \begin{pmatrix} 0 & 1 \end{pmatrix},$$

$$A^{[2]0} = \begin{pmatrix} 0 & 0 & -1 \\ -\frac{1}{\sqrt{2}} & 0 & 0 \end{pmatrix}, \qquad A^{[2]1} = \begin{pmatrix} -\frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & -1 & 0 \end{pmatrix},$$

$$A^{[3]0} = \begin{pmatrix} 0 & -\sqrt{\frac{2}{3}} \\ -\frac{1}{\sqrt{3}} & 0 \\ 0 & 0 \end{pmatrix}, \qquad A^{[3]1} = \begin{pmatrix} -\sqrt{\frac{2}{3}} & 0 \\ 0 & 0 \\ 0 & -\frac{1}{\sqrt{3}} \end{pmatrix},$$

$$A^{[4]0} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}, \qquad A^{[4]1} = \begin{pmatrix} 0 \\ \frac{1}{\sqrt{2}} \end{pmatrix}.$$

The state is then

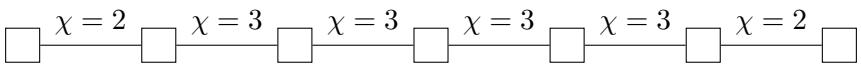
$$|\psi\rangle = \sum_{s_1 s_2 s_3 s_4} A^{[1]s_1} A^{[2]s_2} A^{[3]s_3} A^{[4]s_4} \, |s_1 s_2 s_3 s_4\rangle.$$

### (iii) Bond dimension

$\chi = 3$ for the middle bond. This means that there are 3 possible states for the 2 sites to the left of the bond, since we only have 2 pairs of fermions:

- State 1: both sites occupied.
- State 2: no sites occupied.
- State 3: one site occupied.

These are 3 possible bond states. But one important thing to notice is that if we increase $L$ to 7, all four bonds will still have $\chi = 3$, because we are restricted to 2 pairs. This means that, for any number of sites $L$, the bond dimensions are




So the shapes of the $A$ matrices are $(\chi_{i-1}, \chi_i)$:

$$\begin{aligned} A^{[1]s_1} &: (1 \times 2), \\ A^{[2]s_2} &: (2 \times 3), \\ A^{[3]s_3}, \ldots, A^{[L-2]s_{L-2}} &: (3 \times 3), \\ A^{[L-1]s_{L-1}} &: (3 \times 2), \\ A^{[L]s_L} &: (2 \times 1). \end{aligned}$$

**Constructing the middle $A^{[i]s_i}$.** Use bond indices $\alpha_i \in \{0, 1, 2\}$, which tell us how many sites to the left are already occupied. $A^{[i]s_i}_{\alpha_{i-1},\alpha_i}$ is a $3 \times 3$ matrix with entries labelled by $\alpha_{i-1}, \alpha_i$.

For $s_i = 0$ no new site is occupied, so $A^{[i]0}_{\alpha_{i-1},\alpha_i}$ has entries $\alpha_{i-1} = \alpha_i$ equal to 1 and $\alpha_{i-1} \neq \alpha_i$ equal to 0:

$$A^{[i]0} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

For $s_i = 1$ one more site becomes occupied, so $A^{[i]1}_{\alpha_{i-1},\alpha_i}$ has entries $\alpha_{i-1} = \alpha_i - 1$ equal to 1 and all others equal to 0:

$$A^{[i]1} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

**Unused bond states near the edges.** Close to the edges, not all three bond states are possible. The bond index $\alpha_i$ counts how many pairs sit on sites $1, \ldots, i$. It can never be larger than the number of sites to its left, $\alpha_i \leq i$, and it must leave room on the remaining $L - i$ sites for the $2 - \alpha_i$ missing pairs, $2 - \alpha_i \leq L - i$. This gives

| | |
|---|---|
| $\alpha_0 \in \{0\}$ | nothing is to the left of site 1, |
| $\alpha_1 \in \{0, 1\}$ | one site holds at most one pair, so $\alpha_1 = 2$ is never used, |
| $\alpha_{L-1} \in \{1, 2\}$ | the last site holds at most one pair, so $\alpha_{L-1} = 0$ is never used, |
| $\alpha_L \in \{2\}$ | both pairs have been placed. |

For $2 \leq i \leq L - 2$ all three values $\alpha_i \in \{0, 1, 2\}$ are possible. This is why the matrices shrink to $(1 \times 2)$ and $(2 \times 3)$ at the left edge and to $(3 \times 2)$ and $(2 \times 1)$ at the right edge, and why the bond dimensions are $\chi_i = 1, 2, 3, \ldots, 3, 2, 1$.

## Task 8: From FCI via Hartree-Fock to the DMRG

### (i)

The results below were computed numerically using the code available on GitHub. Figure 3 shows the entanglement entropy of the middle cut as a function of the coupling strength $g \in [-1, 1]$ for $\xi = 1$. It also shows the case where the one-body term is absent ($\xi = 0$), for $g \in [0, 1]$.

For $\xi = 0$ and $g > 0$, the entropy remains constant. As shown in Task 7, the ground state is an equal superposition of all six configurations containing two occupied levels. Consequently,

the Schmidt coefficients across the middle cut remain constant, giving an entanglement entropy of approximately $S = 0.868$.

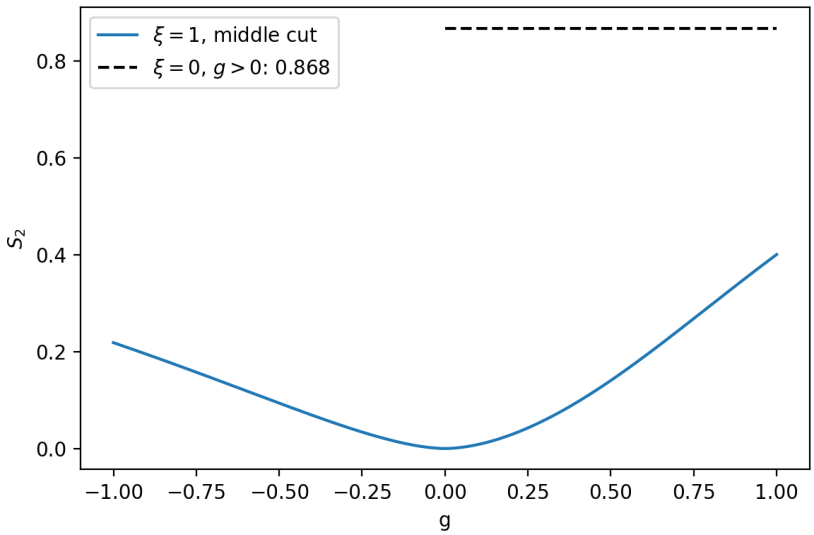



Figure 3: Entropy as function of $g$ for the middle cut with bond dimension $\chi = 3$ and $\xi = 1$. The above line shows the case when $g > 0$ and $\xi = 0$. (GitHub).

When $\xi = 1$, the ground-state eigenvector changes with $g$. As $g \to 0$, the interaction terms vanish and the Hamiltonian becomes diagonal. The ground state is then the configuration $|12\rangle$, which is separable across every cut. Consequently, each cut has a single nonzero Schmidt coefficient equal to one, giving zero entanglement entropy, as shown in Figure 4.

As $g$ moves away from zero, other configurations contribute to the ground state, changing the Schmidt coefficients and increasing the entanglement entropy. The middle cut can support greater entanglement than the outer cuts because its maximum bond dimension is $\chi_2 = 3$, compared with $\chi_1 = \chi_3 = 2$. This allows up to three nonzero Schmidt coefficients across the middle cut, whereas the outer cuts allow at most two.

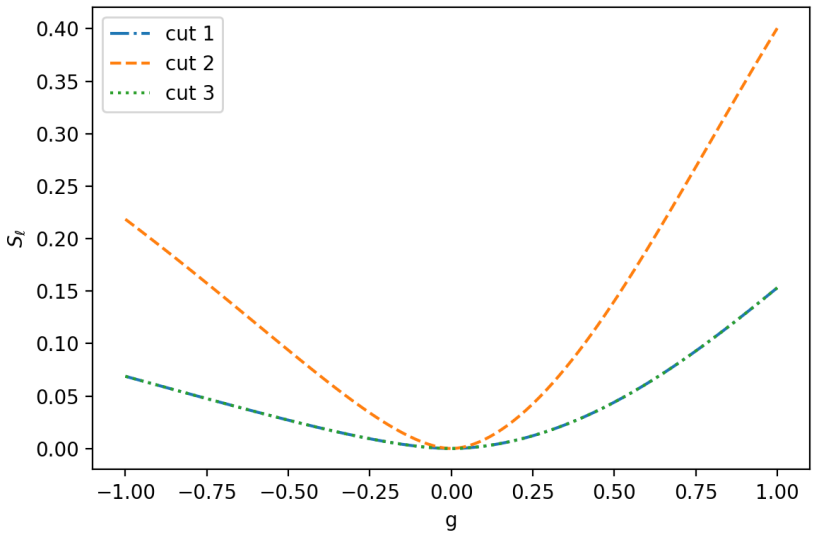



Figure 4: Entropy as function of $g$ for all 3 cuts. the cuts have bond dimention $\chi = [2, 3, 2]$ (GitHub).

**(ii)**

**Hartree–Fock state as an MPS.** The Hartree–Fock orbitals of Task 4 self-consistent for all $g$, so the Slater determinant is the basis state $|\Phi_{\mathrm{HF}}\rangle = |1100\rangle$, with coefficient vector $(1,0,0,0,0,0)$. The coefficient matrix will only have 1 entry with amplitude 1 across the three cuts, so the Hartree–Fock state is a product state in the basis $|n_1 n_2 n_3 n_4\rangle$ with bond dimensions $(1,1,1)$.

**Truncation across the middle cut.** The results used in this task were computed with the script GitHub. The coefficient matrix across the middle cut is $4 \times 4$. Its SVD gives the Schmidt weights $\sigma_k^2$. Keeping the $\chi$ largest singular values and renormalising gives the truncated state, while the discarded weight is $\sum_{k>\chi} \sigma_k^2$.

For $\chi = 1$, the dominant Schmidt state is $|1100\rangle$ for all $g \in [-1,1]$, giving $E_{\chi=1} = 2 - g$, equal to the Hartree–Fock energy. Increasing $\chi$ to 2 greatly reduces the discarded weight and energy error, showing that one additional Schmidt state captures most of the missing correlation energy. At $g = 0$, the exact ground state is $|1100\rangle$, so all approximations are exact. The errors increase with $|g|$ and are larger for positive $g$, consistent with the greater entanglement.

The truncated CI of Task 3 gives slightly lower energies than $\chi = 2$, because it minimises the energy within its chosen subspace, whereas the SVD truncation optimises the overlap with the exact state. All approximate energies lie above the exact ground-state energy, as required by the variational principle. The ordering is

$$E_{\mathrm{exact}} < E_{\text{trunc. CI}} \lesssim E_{\chi=2} \ll E_{\chi=1} = E_{\mathrm{HF}}.$$

**Two kinds of best product state.** Hartree–Fock finds the Slater determinant with the lowest energy, while $\chi = 1$ finds the product state with the largest overlap with the exact state. These are different optimisation criteria and need not give the same state.

In this model, they coincide: the Hartree–Fock state is $|1100\rangle$, which is also the dominant Schmidt state throughout the considered range of $g$. Both therefore give the energy $2 - g$. This agreement is specific to this model and is not guaranteed in general.

| $g$ | $E_{\mathrm{exact}}$ | $E_{\mathrm{HF}} = E_{\chi=1}$ | $E_{\chi=2}$ | $E_{\text{trunc. CI}}$ | $\sum_{k>1} \sigma_k^2$ | $\sum_{k>2} \sigma_k^2$ |
|---|---|---|---|---|---|---|
| $-1$ | 2.77987 | 3.00000 | 2.78696 | 2.78531 | $5.5 \cdot 10^{-2}$ | $1.0 \cdot 10^{-3}$ |
| $-0.5$ | 2.43688 | 2.50000 | 2.43775 | 2.43737 | $1.9 \cdot 10^{-2}$ | $1.4 \cdot 10^{-4}$ |
| 0.5 | 1.41677 | 1.50000 | 1.41785 | 1.41760 | $3.1 \cdot 10^{-2}$ | $1.5 \cdot 10^{-4}$ |
| 1 | 0.63555 | 1.00000 | 0.64989 | 0.64891 | $1.3 \cdot 10^{-1}$ | $1.7 \cdot 10^{-3}$ |

Table 1: Energies for $\xi = 1$: exact ground state (Task 2), Hartree–Fock (Task 4), which coincides with the $\chi = 1$ truncation of the exact state, the $\chi = 2$ truncation across the middle cut, and the truncated CI of Task 3 (reference plus four single-pair excitations). The last two columns give the discarded Schmidt weight of the middle cut for $\chi = 1$ and $\chi = 2$ (GitHub).

## (iii) MPO check and energies of the truncated MPS

The results used in this task were computed with the script GitHub. The MPO from Task 5 was contracted to obtain the full $16 \times 16$ Hamiltonian matrix. Restricted to the six two-pair states, it agrees exactly with the Hamiltonian from Task 2, confirming that the MPO is implemented correctly. The full matrix is also Hermitian and conserves the number of pairs, as expected.

The energies of the truncated MPS were then calculated by contracting the MPO with the MPS. They agree with the results from (ii), confirming that both methods give consistent energies. Table 2 shows selected results.

As expected, increasing $\chi$ improves the energy by retaining more Schmidt states. For $\chi = 4$, no states are discarded, and the exact energy is recovered. For $\chi = 1$, the MPS is the product

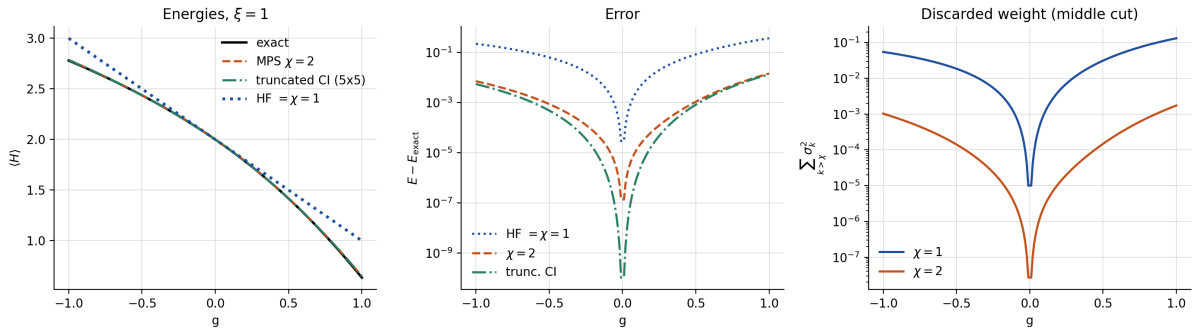



Figure 5: Truncation of the exact ground state across the middle cut for $\xi = 1$ as a function of $g$. Left: energies of the exact state, the $\chi = 2$ truncation, the truncated CI of Task 3, and the Hartree–Fock energy, which coincides with the $\chi = 1$ truncation. Middle: the error $E - E_{\text{exact}}$ on a logarithmic scale. Right: the discarded Schmidt weight $\sum_{k>\chi} \sigma_k^2$ for $\chi = 1$ and $\chi = 2$ (GitHub).

| $g$ | $\chi$ | discarded weight | $\langle \hat{H} \rangle$ (MPO) | $E_{\text{exact}}$ |
|---|---|---|---|---|
| $-1$ | 1 | $5.5 \cdot 10^{-2}$ | 3.00000000 | 2.77987014 |
| $-1$ | 2 | $1.0 \cdot 10^{-3}$ | 2.78696058 | 2.77987014 |
| 1 | 1 | $1.3 \cdot 10^{-1}$ | 1.00000000 | 0.63554847 |
| 1 | 2 | $1.7 \cdot 10^{-3}$ | 0.64989429 | 0.63554847 |
| 1 | 4 | 0 | 0.63554847 | 0.63554847 |

Table 2: Truncated MPS energies calculated using the MPO, compared with the exact ground-state energies. GitHub.

state $|1100\rangle$, giving the Hartree–Fock energy $E = 2 - g$. All truncated energies are above or equal to the exact ground-state energy, consistent with the variational principle.